



Physics-guided spatiotemporal deep learning for early melt pool instability detection in dual-beam laser welding of steel–titanium joints

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ABSTRACT

The concept of dual-beam laser welding is the new method of welding dissimilar materials like steel and titanium, with a better process stability and quality of the joint, in contrast with single-beam methods. This paper presents a deep learning-based model to classify melt pool instabilities during high-speed thermal imaging of steel-titanium joints of high-speed laser welding. Infrared image sequences with a high frame-rate of changing melt pool dynamics are obtained over a variety of conditions of the process. Spatiotemporal deep learning architecture that is based on convolutional neural networks to extract spatial features and recurrent temporal models learning is utilized to learn discriminative thermal patterns of various instability modes. The proposed model is trained with a categorical cross-entropy criterion and is tested with extensive measures, such as accuracy, F1-score, and receiver operating characteristic (ROC)-area under the curve (AUC). The experimental findings indicate strong classification and all instability classes with the F1-scores of over 88% and the AUC of over 0.90. Absorption-informed feature inclusion gives significant increases in first instability detection and general discriminative ability over baseline feature representations. The results confirm that deep learning using high-speed thermal imaging can be effective to model the high-level spatiotemporal signatures of the melt pool instabilities and this forms a promising direction to real-time monitoring and closed-loop control in advanced laser welding systems.

1. Introduction

The superior capability of dual-beam laser welding to redistribute energy input, stabilize keyhole behavior and enhance the integrity of the joint over and above the conventional single-beam systems has led to a rise in its use as an advanced joining method of dissimilar materials, especially steel-titanium combinations [1]. In high-performing industries like aerospace, automotive and energy, steel-titanium joints provide a good balance of weight, corrosion, and strength [2]. Nevertheless, large thermalphysical differences between steel and titanium

such as thermal conductivity, absorptivity, melting temperature, and heat capacity make asymmetric heat transfer and dynamic melt pool behavior during welding [3]. Recently, dual-beam designs, particularly offset or coaxial beam designs have been demonstrated (2022–2025) to improve thermal distributions as well as to reduce extreme spatter formation and enhance penetration stability. However, when two beams that are controlled independently interact, this further complicates the melt pool in space and time and requires a more complex method of monitoring and analysis.

Even with the benefits of the dual-beam systems, the instability of the

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melt pool is one of the issues that are most critical in attaining the quality and repeatability of the welds in dissimilar metal welding. Direct effects of instability modes, including keyhole collapse, humping, ejection of spatter, partial fusion, formation of porosity, and over-penetration, directly affect the geometry of a weld, evolution of a weld microstructure, and mechanical stabilization in the long run [4–7]. Overlapping thermal fields and intermittent changes in energy absorption in dual-beam welding enhance transient oscillatory behaviour on the molten pool. It has been recently established in recent studies (2022–2025) that whereas dual-beam strategies can reduce certain instability mechanisms, they equally lead to highly dynamic melt pool behavior which cannot be described in terms of traditional physical observation techniques. Thus, intelligent welding and adaptive control require detecting and classifying instability events in the near future and reliably.

Traditional monitoring instruments do not have the power to address in detail the intricate spatiotemporal development of melt pool instabilities. Photodiodes and pyrometers are sensors that can only give thermal measurements at a point or average values but not a spatial resolution [8,9]. Image processing algorithms that are based on rules are sensitive to changes in emissivity, noise, and processes, rely on hand-crafted features, and predefined thresholds [10]. Although currently high-speed infrared thermography has also made it possible to capture melt pool temperature field at the kilohertz level, which allows the identification of rapid keyhole oscillations and spatter initiation events [11,12], due to the high dimensionality and time-dependent nature of the acquired data, computational models that have higher abilities to extract meaningful patterns that cannot be easily interpreted by the human eye are required.

Simultaneously, deep learning has shown impressive abilities in the modeling of nonlinear relationships in sequence image series of high dimensions. Newer convolutional neural networks (CNNs) have been applied with a temporal convolutional neural network, e.g. long short-term memory (LSTM) and gated recurrent units (GRUs), which have delivered better results in defect detection and predicting process stability (2022–2025 studies). These methods are superior to the conventional threshold-based methods which learn discriminative spatiotemporal characteristics directly of data. Nevertheless, the literature is mostly conducted on single-beam welding or on a homogeneous material and very little research has been conducted on the dual-beam laser welding of dissimilar steel-titanium joints using high-speed thermal imaging. Moreover, the combination of physics-guided feature augmentation with spatiotemporal deep learning has not been fully studied yet.

It is on these challenges and opportunities that this study suggests a deep learning-based framework of the classification of melt pool instabilities in the dual-beam laser welding of steel-titanium joints using high-speed thermal imaging. The main contributions made by this work are summarized as follows:

- ✓ Developed a spatiotemporal deep learning framework achieving high classification accuracy for melt pool instabilities in dual-beam laser welding.
- ✓ Demonstrated that high-speed thermal imaging effectively captures distinctive spatiotemporal signatures of keyhole, humping, spatter, and fusion-related instabilities.
- ✓ Established that physics-guided feature augmentation improves ROC-AUC and early instability detection compared to baseline thermal features.
- ✓ Validated real-time feasibility of the proposed model, achieving low-latency inference suitable for industrial weld monitoring and adaptive control systems.

Collectively, these contributions help the state of the art in intelligent laser welding, as they make it possible to identify the instabilities of the melt pool with the help of accurate, data-driven and physically

consistent data, which opens the door to real-time quality assurance and adaptive process control.

Nomenclature used

CNN – Convolutional Neural Network
 LSTM – Long Short-Term Memory
 GRU – Gated Recurrent Unit
 ROI – Region of Interest
 ROC – Receiver Operating Characteristic
 AUC – Area Under Curve
 IR – Infrared
 α – Material absorption coefficient
 P – Laser power
 V – Welding speed
 ρ – Density
 Cp – Specific heat
 k – Thermal conductivity
 Tm – Melting temperature

2. Related work

2.1. Melt pool monitoring in laser welding

The use of melt pool monitoring methods to learn about the stability of the process and the quality of weldings in laser welding was widely researched. The characteristics of plasma radiation and vapor plume were extensively studied by means of optical emission spectroscopy to be able to indirectly determine the keyhole stability and penetration behavior [13]. Though these techniques were effective in identifying gross deviation in the processes, they were very sensitive to shielding gas composition and plume variation thus minimizing reliability even under challenging welding conditions [14].

Other researchers had also introduced coaxial vision systems in which the optical path of the laser was combined with visible or near infrared cameras to watch melt pool geometry and surface dynamics in real-time [15]. Such systems were able to detect the defects of humping and excessive penetration [16]. Nevertheless, they were negatively affected by high process radiation, spatter blockage, and surface reflectivity disparity, which was exerted in dissimilar material welding [17]. Infrared thermography was also developed as a viable alternative because it gave spatially resolved temperature information on the melt pool and its heat-affected zone [18]. Transient thermal gradients, oscillatory behavior and cooling rates that were captured using high-speed thermal imaging closely correlated with melt pool stability [19]. Irrespective of these benefits, majority of the research on thermography involved single-material or single-beam welding. There was a paucity of studies involving dissimilar material systems, e.g. steel-titanium joints, and especially in the consideration of emissivity, absorptivity, and thermal conductivity mismatches [20].

Melt pool stability can only be accurately monitored by a detailed characterization of transient heat distribution patterns and spatial temperature gradient characterizations. Thermal gradients in the molten body regulate the behavior of fluid flow, Marangoni convection and the development of solidification fronts, which in turn directly determine the instability formation. Other more recent developments (2022–2025) have shown that steep temperature gradients at the keyhole edge are highly related to keyhole oscillation frequency and spatter start. Since in dual-beam systems the heat sources of two lasers are superimposed, there are asymmetrical and time-varying thermal fields which cause local overheating or uneven solidification. Parameters that can be quantified by the use of high-resolution thermography include max temperature, gradient magnitude (degree centigrade/millimeter), and rate of cooling (degree centigrade/milliseconds), which are discriminatory parameters of instability modes. Thus, spatially resolved thermal gradient analysis is needed to be able to separate stable melt pools and dynamically unstable regimes.

In spite of its benefits, emissivity variation, surface oxidation and

dynamic phase change are some of the challenges that face infrared thermography in laser welding. The emissivity of molten metal also varies with that of solidified areas and can be highly different between steel and titanium as a source of uncertainty in measuring absolute temperature. Also, high-energy welding can create distortions in thermal signals due to the formation of plasma as well as vaporization of metals. Recent studies have highlighted that a camera calibration, emissivity compensations model and shielding must be used to enhance the reliability of measurements. With the dual-beam welding of dissimilar materials, emissivity mismatch in the interface of both welds also complicates the thermal interpretation of the interface. Thus, strong monitoring strategies should consider these variations by either correcting them using some form of calibration or modeling the variations using the data with the ability to learn invariant thermal characteristics.

The composition and flow rate of shielding gas is also very important to the stability of the melt pool and fidelity of monitoring. There are many effects of the use of argon shielding; it inhibits oxidation, it affects the plume dynamics and suppresses spatter, which also affects the visible thermal signature. The change in convective heat transfer concerning variation of gas flow can alter the surface emissivity as well, affecting the validity of thermal images. Also, there is no overestimation in the significance of high frame-rate acquisition. Keyhole collapse and spatter ejection are a few milliseconds in duration; therefore, to record oscillatory events fast enough to eliminate time aliasing, kilohertz level imaging is needed. Systems with low frame-rates can fail to detect important transitional states and their accuracy is lower. Thus, to fully realize reliable monitoring of melt pool instability in dual beam laser welding machines, it is essential to integrate high-speed thermography with highly advanced spatiotemporal analysis (Table 1).

2.2. AI and Deep Learning in Welding

As sensing and computing technology improved, researchers used artificial intelligence methods more often to monitor the welding process. CNNs were able to identify defects in weld using optical images, which allowed identifying porosity, cracks, and underfill defects with high accuracy and reliability [21]. As these methods proved to be effective in extracting spatial features, they generally worked at the frame-by-frame level and did not consider the temporal changes of melt pool dynamics. To compute temporal effects, researchers used recurrent neural network models, including long short-term memory (LSTM) and gated recurrent unit (GRU) models to analyze stability in welding and predict quality [22]. These models enhanced the response to the short-term process variations and initial defect formation on a pattern of learning [23]. Nevertheless, most studies used low-dimensional sensor signals or hand-designed features instead of high-rate sequences of thermal images and thus could not capture intricate spatiotemporal processes.

Another study delved into transfer learning to solve the problem of the scarcity of data in manufacturing vision by finetuning the existing convolutional backbones to welding tasks [24]. Though the strategy

enhanced model convergence, generalization, majority of the transfer learning-based researches focused on single-beam welding configurations and homogenous materials. Therefore, there were still significant gaps in the research, such as the absence of deep-learning-based frameworks to dual-beam laser welding, the insufficient use of high-speed thermal sequence modeling, or the lack of research based on the classification of melt pool instability in steel and titanium dissimilar joints. The current paper was inspired by these constraints (Table 2).

3. Experimental setup and data acquisition framework

3.1. Dual-beam laser welding setup

This experiment used a two-beam laser welding system in order to provide reproducibility of the experiment and controlled study of the melt pool instabilities in steel-titanium dissimilar joints. Dual-beam welding configurations were earlier demonstrated to be more effective in terms of energy distribution and stability of the process than in single-beam welding, especially with dissimilar material systems [24]. Laser system had two beams which were independently controllable thus beam power, wavelength and spatial separation could be controlled very precisely. The main beam provided enough energy to cause the process of melting and stabilization of the keyhole, and the secondary beam transferred heat to control the geometry in the melt pool and eliminate large variations, as is expected with reported dual-beam interaction processes [25].

The experiments have used steel and titanium alloys which have vastly different thermalphysical characteristics such as thermal conductivity, melting temperature, and absorptivity. Earlier researchers pointed out that such property mismatches have a potent impact on heat flow asymmetry and stability of melt pools in dissimilar welding [26]. Steel had a higher ability to conduct heat off the interaction zone compared to titanium which conducted heat locally and created asymmetric heat gradients across the joint interface. The experimental took the shape of a butt-joint design to provide uniformity in the distribution of heat and the ability to recreate the fusion behaviour. The butt joints were frequently utilized in studies on laser welding to reduce the uncertainty in geometry and also to make the observation of the melt pool simpler [27]. The exact positioning and firm clamping ensured that there were no changes in gap conditions and avoided mechanical distortion in the process of welding.

The molten metal was shielded against oxidation and contamination in the atmosphere by argon shielding gas. Stable flow of the shielding gas controlled the surface of the melt pool and minimized the spatter formation, as previously determined in the welding research on lasers [28]. The rate of welding was controlled in a systematic manner to control linear heat input and cooling rate that directly affected the stability of the melt pool and solidification behavior [29]. The Fig. 1 represents the dual-beam welding system applied in the welding of steel and titanium in butt-joint. The joint area is exposed to two laser beams that are traversed separately and argon shielding gas protects the molten

Table 1
Summary of related work on melt pool monitoring in laser welding.

Monitoring Technique	Sensing Modality	Information Captured	Strengths	Key Limitations	Applicability to Dissimilar Welding
Optical Emission Spectroscopy	Plasma radiation	Keyhole stability, penetration trends	High temporal sensitivity	Sensitive to plume fluctuation and shielding gas	Limited
Coaxial Vision Systems	Visible / NIR imaging	Melt pool shape and surface dynamics	Real-time visualization	Affected by spatter and intense radiation	Moderate
Infrared Thermography	Thermal imaging	Temperature field, gradients, cooling rates	Spatially resolved thermal data	Emissivity and calibration challenges	High
High-Speed Thermal Imaging	IR thermography (kHz)	Transient melt pool dynamics	Captures spatiotemporal behavior	Large data volume	High
Rule-Based Image Analysis	Threshold-based features	Geometric indicators	Simple implementation	Poor robustness to noise	Low

Table 2
Summary of related work on AI and deep learning in welding.

Approach	Input Data Type	Learning Capability	Main Application	Key Limitation	Research Gap
CNN-Based Models	Optical images	Spatial feature learning	Weld defect detection	Ignores temporal evolution	No instability dynamics
LSTM / GRU Models	Sensor time-series	Temporal dependency modeling	Weld stability prediction	Limited spatial representation	No thermal imaging
CNN-RNN Hybrids	Image sequences	Spatiotemporal learning	Quality monitoring	Mostly single-beam studies	No dual-beam focus
Transfer Learning	Pretrained vision models	Improved generalization	Manufacturing inspection	Domain mismatch	Limited thermal adaptation
Traditional ML	Handcrafted features	Shallow nonlinear mapping	Geometry prediction	Feature dependency	Poor scalability

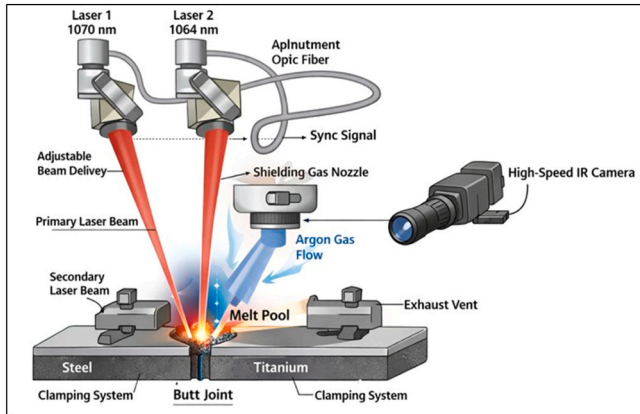


Fig. 1. Experimental Setup for dual-beam laser welding of steel–titanium dissimilar joints with high-speed thermal monitoring.

pool. The thermal behavior of the melt pool is captured in real time at a high speed infrared camera to allow the analysis of stability and instability behaviour under conditions of controlled process scenarios.

The experimental design was arranged in such a way that it provided stable and repeatable conditions of dual-beam laser welding, set up represent in Fig. 2. The fibre laser system we used has a wavelength of 1070 nm, which has been extensively applied in the high-precision processing of metals because of its constant quality of the beam and highly effective absorption in steel and titanium. Beam-1 and Beam-2 used a 0.6–0.8 mm and 0.5–0.7 mm focal length respectively, and a beam profiler was used to stabilize the energy density of the beam. The ratio of beam overlap was set between 40 and 60 in order to examine the interaction effects and stability of the melt pools. The 35 mm to 5 mm



Fig. 2. Experimental dual Beam laser welding setup representation.

spectral range was used as the high-speed infrared imaging range because it is in the range of peak thermal emission at that time of welding. A calibrated IR camera was used to observe the application of this spectrum. Radiometric calibration of the radiometric was performed on a blackbody reference source at 300–1200 °C range before each experimental run. Thermal data were recorded at a sampling rate of 2–5 kHz to record the fast fluctuation events. The overall temperature reading error was not more than 2 % as per the emissivity modification and the calibration check.

3.2. High-speed thermal imaging system

The researchers used a high-speed infrared thermal imaging device to record dynamic flows of a transient melt pool when welding using a dual beam laser. High-speed thermography has been shown to be an effective instrument in tracking dynamic thermal behavior e.g. keyhole oscillations, melt pool instability, etc. [30]. The infrared camera had filming rates in the kilohertz frequencies and a mid-wave infrared spectrum that was applicable to high-temperature metal processing purposes. The laser source was in sync with the imaging system so that the laser and material interaction and thermal data acquisition were time matched. To correlate process events with the thermal response, synchronization was necessary and was widely used in the improved diagnostics of welding [31]. The camera was also calibrated to the temperature to remove the change in emissivity and sensor nonlinearity so that the infrared radiations could accurately estimate the temperature [32].

The area of interest (ROI) chosen in the study was at the melt pool and around the area that was affected by heat. The ROI-based analysis minimized the background interference and the computing expenses and retained the thermally relevant data, which was in accordance with the previous findings of thermal imaging in welding [33]. ROI was always able to capture the keyhole, melt pool boundaries and solids front. Patterns of thermal images showed different patterns in the case of stable and unstable welding. The melt pools that were stable displayed the continuous temperature distributions and slow temporal change, and the unstable ones demonstrated sharp temperature peaks and the oscillations, as the previous thermographic studies of the welding processes revealed [34]. Fig. 3 demonstrates that high-speed infrared thermography can be used to measure transient melt pool behaviour when welding with a dual-beam laser. The system records the rapid changes in thermal variation in an area of interest that is focused around the keyhole, melt pool, and neighboring heat affected zone. With the condition of steady welding, smooth temperature gradient with time and uniform geometry of the melt pool are observed in the thermal images. On the contrary, unstable cases are described by the localized temperature rises, localized melting pool boundaries and the strong time oscillations. This stark contrast of imaging clearly shows the potential of synchronized, calibrated thermal imaging to be used in the differentiation of stable and unstable process regimes which allows the dynamic of the melt pool and process stability to be diagnosed reliably (Fig. 4).

3.3. Melt pool instability taxonomy

The paper has created a melt pool instability taxonomy that determines the consistent classification labels of deep learning-based

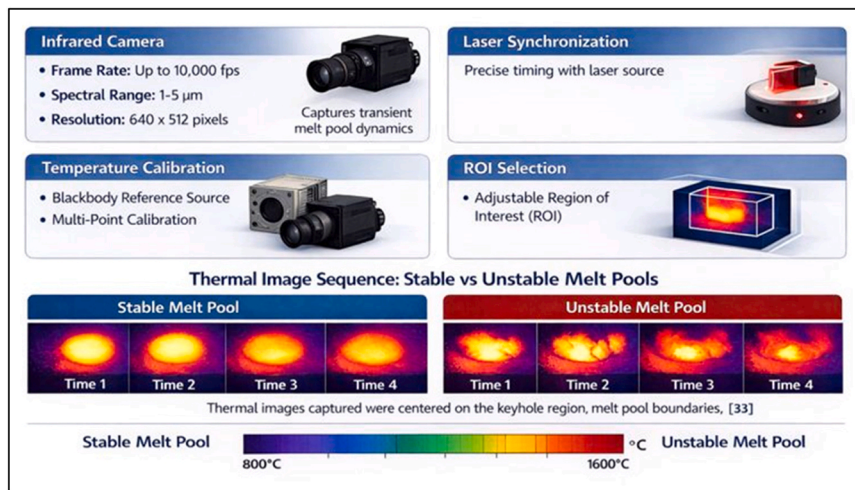


Fig. 3. High-Speed Infrared Thermography for Melt Pool Stability Analysis in Dual-Beam Laser Welding.



Fig. 4. High Speed camera device for data capture and experiment system.

analysis. The steady melt pools were found to be related to uniform, tranquil thermal fields with very little temporal variation which were previously related in the literature to uniform keyhole behavior and homogeneous penetration [35]. Keyhole instability happened at times where vapor cavity oscillation or collapse was a result of higher energy input. The earlier studies attributed such behavior to the high-frequency thermal fluctuations around the keyhole orifice [36]. Instability at the humping presented itself as periodic deposition of molten material on the weld seam, or generated cyclic thermal peaks and long hot zones that corresponded to the direction of the welding [37].

Violent instability was caused by melt pool agitation and ejection of droplets that led to spatter. These events were detected in the thermal imaging investigation as sudden bursts of local temperature and abnormal hot spots trajectories [38]. Incomplete fusion or incomplete non-fusion was observed with either a low input of heat or too fast welding rate resulting in lower peak temperatures and even less thermal coverage at the joint interface [39]. Over penetration occurred when

heat input was greater than optimal resulting in sustained high temperature and increased thermal regions above the desired fusion area. These conditions had been in the past associated with a deep keyhole formation and erratic melt pool behaviour [40]. This taxonomy gave a purposeful and physically explainable foundation of labeling thermal data that was used to prepare and test the suggested deep learning classification framework. Summarizing the different classes of melt pool instability by relating physical definitions thereto with their typical thermal signatures gives a clean and consistent foundation on which high-speed thermal data that can be used as inputs in deep learning-based classification is labeled, as shown in the Table 3.

3.4. Justification of process parameter selection

The chosen range of beam power (9001300 W in the case of Beam-1 and 7001100 W in the case of Beam-2) was chosen due to recent research on dual-beam laser welding (20232025) that showed consistent penetration and controllable development of a keyhole within similar energy densities in steel-titanium and other dissimilar systems. The welding rates (4.5–6.0 mm/s) were also systematically manipulated to cover both stable and unstable regimes, allowing the smooth passage to the keyhole oscillation, humping and excessive penetration. The power-to-speed ratios were used to compute linear heat input (267533 J/mm) on purpose to achieve various instability classes to generate sound datasets. The choice of the shielding gas (argon, 1520 L/min) was based on the principles of the existing industry practice to ensure the lowest levels of oxidation and stabilize the melt surface and the low level of spatter, keeping the thermal conditions constant.

3.5. Material selection rationale

In this study, steel-titanium dissimilar joints were chosen because they are increasingly becoming relevant in the industry in lightweight, high-strength structural. Titanium alloys find extensive applications in the aerospace systems due to their high-strength to weight ratio, corrosion and high-temperature stability, whereas advanced steels offer cost-effective structural reinforcement and high-load bearing strength. The steel-titanium hybrid assemblies allow optimization in the structural performance through the design of resourceful combination of material benefits without adding much weight or making it more expensive to make. Likewise, in biomedical and energy applications titanium is needed due to its biocompatibility and corrosiveness and steel is required due to its mechanical strength and structural integrity. Such materials are thus becoming more required in the next generation transportation platforms, pressure vessels and in the hybrid structural

Table 3
Melt pool instability taxonomy and corresponding thermal signature characteristics.

Instability Type	Definition	Key Thermal Signature Characteristics	Physical Interpretation
Stable Melt Pool	Welding condition with steady melt pool geometry and uniform fusion	Smooth temperature distribution, symmetric thermal profile, low temporal fluctuation	Balanced heat input and stable fluid flow
Keyhole Instability	Unstable vapor cavity formation causing oscillation or collapse	High-frequency temperature oscillations, abrupt intensity variations near keyhole region	Excessive or fluctuating energy input
Humping	Periodic accumulation of molten material along weld seam	Cyclic thermal peaks, elongated hot regions propagating along welding direction	Unstable melt flow and solidification imbalance
Spatter-Dominated Instability	Ejection of molten droplets from melt pool	Sudden localized temperature spikes, irregular moving hot spots	Violent melt pool agitation
Partial / Lack of Fusion	Incomplete melting at joint interface	Reduced peak temperatures, discontinuous thermal coverage at joint line	Insufficient heat input or excessive welding speed
Excessive Penetration	Over-penetration beyond optimal fusion depth	Sustained high temperatures, expanded thermal footprint below joint	Excessive heat input and deep keyhole formation

components.

Nevertheless, the welding of titanium and steel result in serious metallurgical and thermal difficulties associated with the strong incompatibility of the thermophysical properties. Titanium has a low thermal conductivity or thermal conductivity with a high melting temperature relative to most of the steels, and creates an asymmetric heat flow and varying cooling rates during welding. This is more likely to benefit intermetallic compound development at the interface, increase thermal gradients and susceptibility to interfaces melt pool instabilities including keyhole fluctuation and partial fusion. Therefore, steel -titanium welding offers a perfect example to study complicated melt pool behavior in dual-beam laser conditions.

3.6. Experimental procedure

To start the experimentation, we initially prepared the material surfaces so that the quality of weld is uniform. The steel and titanium samples were mechanically polished with a series of abrasive documents of successively finer grits (to 1000 grit) to eliminate oxides and other external contaminants. The samples were then washed using acetone to remove any grease and dust then dried them using compressed air.

The next it placed the specimens in a rigid clamping hold on to ensure that the butt-joints were properly aligned and distortion reduced during welding. To ensure that there were no artificial penetration variations, and checked alignment with a digital vernier caliper and made sure that there was a zero-gap interface.

Both laser beams were then calibrated. Study opted to use a beam profiler and an alignment target by focusing on the position and checking the diameter of the spot (range of 0.6–0.8 mm). The ratio of predetermined beam overlap was as well adjusted and coaxial alignment

was ensured to provide consistency of energy interaction.

Experimentation measured the infrared camera with a blackbody reference source in 3001200 sous C before welding. The values of emissivity are determined based on conditions of the materials and radiometric accuracy.

The study also did the welding runs on controlled argon shields of 1520 L/min. The constant travel speed and power settings were held during every run and high-speed thermal sequences were recorded at 25–50 Hz.

Later it also visually examined the weld bead after welding to identify surface defects and reported macro-features. A precision cutter was used to section selected samples across-the-welding line, fix the samples in epoxy, polish the samples, and etch the samples to cross-sectional examine the penetration depth and defect morphology.

4. Methodology

Fig. 5 demonstrates the general flow of the proposed melt pool instability classification system that will be created in the context of dual-beam laser welding steel-titanium joints. It starts with the dual-beam welding step: two laser beams are brought to an independent control, and a contact of dissimilar metal interface is brought under the shield of argon, forming a dynamic melt pool with complicated thermal and fluid dynamics. These highly dynamic thermal processes are monitored with a high-speed infrared thermal imaging system, which runs at kilohertz frame rates and allows one to acquire detailed information on the temporal flows of temperature. The obtained thermal sequences are then processed according to the Region of Interest (ROI) extraction where the melt pool and surrounding heat-affected area are extracted to remove the irrelevant background information and simplify

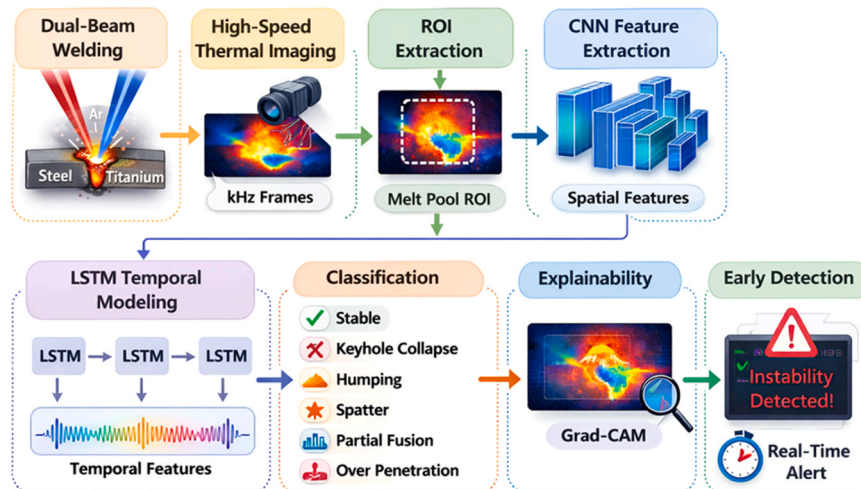


Fig. 5. Overall Workflow of the Proposed Melt Pool Instability Classification Framework.

the processes without the need to calculate all the thermal responses at the same time without significant loss of the thermal characteristic features.

The ROI frames are then preprocessed then transferred to a convolutional neural network (CNN), which extracts spatial features through learning hierarchical representations of the geometry and temperature gradients and localized hot spots of the instability modes. To learn dynamic changes, the spatial descriptions are inputted into a long short-term memory (LSTM) that learns sequential dependence between several consecutive frames so that the system can detect sudden changes, like keyhole collapses and spatter, as well as smooth transitions, such as humping or over-penetration. The CNNLSTM model provides a multi-class prediction of the stable welding or actual instability, such as keyhole collapse, humping, spatter, partial fusion, and over-penetration. An explainability module based on Grad-CAM is used to provide more interpretable results on which areas of the thermal frames contribute the most to classification decisions to gain physically meaningful insights into the behavior of melt pools. Lastly, the classified output is converted to an early detection system, which produces real-time signals of instability development. This end-to-end system allows precise, decodable and low-latency surveillance to be practical in smart welding apparatus and upcoming closed-loop control of the procedure.

4.1. Dataset preparation

The dataset was created based on a sequence of controlled experiments of dual beam laser welding on steel titanium joints in the study. Welds were done two times with different process conditions to obtain both stable and unstable melt pool conditions. High-speed thermal imaging produced huge amounts of thermal frames, which were formatted into fixed length sequences of image that depicted the dynamic behavior of the melt pool over time. To prevent data leakage, the data was split into training, validation, and testing components by the run-wise split method to assure a strong evaluation of performance. Class-weighting methods were implemented when training the models in order to solve class imbalance between various types of instabilities. Such methods of data augmentation as spatial rotation, Gaussian noise injection and temperature scaling improved model generalization and resistance to variability in measurements. The thermal sequences were annotated by domain experts according to the established instability taxonomy so that there is proper labeling. The dataset construction process is described in the Table 4 below and includes runs of weld, structure of thermal frame, data splitting policy, data-augmentation strategy, class imbalance management and labeling of data by experts to provide strong models and robust evaluation.

4.2. Deep learning methodology

4.2.1. Convolutional Neural Network (CNN)

This paper used a convolutional neural network (CNN) as a reference deep-learning architecture in order to train spatial thermal characteristics that became related to melt pool dynamics during dual-beam laser

welding. The CNN was used to process single thermal frames obtained in high-speed sequences of infrared images and the emphasis was put on spatial patterns like shape and size of the melt pool, temperature gradients and hot spots. The features of the edge and thermal boundaries and melt pool geometry and heat distribution were apprehended by convolutional layers with small receptive fields and the higher level features respectively. The layers of pooling dimensionality space and boosted resistance to noise and minor space variations. The CNN-based method did a good job in determining spatial features of the stability and instability of welds, although it considered frames separately and did not explicitly consider how melt pool dynamics changed over time. Consequently, the CNN had a poor performance in separating instability modes sharing similar spatial patterns but unlike temporal mode, e.g., transient keyhole oscillation or progressive humping development. This drawback stimulated the inclusion of the time modelling elements in future architectures.

Let $X_t \in R^{H \times W}$ is CNN extracts spatial features by use of multiple convolutional layers that are specified as follows.

$$F_t^l = \sigma(W^l * F_t^{l-1} + b^l)$$

Where F_t^l refers to the feature map at the l th layer, W^l and b^l are learnable convolution kernels and convolution biases, and $*$ is a convolution operation, and σ is a nonlinear activation.

After spatial pooling:

$$\tilde{F}_t = \text{Pool}(F_t^l)$$

The frame-level prediction is obtained as:

$$\hat{y}_t = \text{Softmax}(W_c \tilde{F}_t + b_c)$$

The formulation processes are independent of each other and it is not capable of models of temporal dependencies between thermal sequences, thus making it inadequate to represent instabilities of melt pools dynamically.

4.2.2. CNN-GRU

The study used a hybrid CNNGRU architecture to integrate spatial feature extractor and sequential learning to incorporate the temporal information. This model operates by first processing the CNN separately on each incoming thermal frame to produce spatial feature vectors which were inputted into a gated recurrent unit (GRU) network to predict temporal recurrence effects with regard to each new frame. GRU was very effective in capturing short- and mid-range time variations like a fast-changing temperature and repetitive thermal cycles, which are typical of some of the melt pool instabilities. It had a gated structure that inhibited retention of irrelevant information over time and reduced the effects of vanishing gradients. The CNNGRU model achieved a better classification than CNN frame based, due to its ability to utilize temporal continuity in the thermal sequence. Nevertheless, the GRU was not quite as sensitive to long-term temporal dependencies because of its comparatively small memory composition, which included a slow change between stable welding conditions and unstable ones. This finding indicated that a more expressive temporal model should be developed.

Spatial features derived through the CNN are arranged in sequence in order to add in the time dynamics.

$$Z_t = \text{CNN}(X_t), \quad t = 1, \dots, T$$

The gated recurrent unit (GRU) models temporal evolution as:

$$z_t = \sigma(W_z Z_t + U_z h_{t-1})$$

$$r_t = \sigma(W_r Z_t + U_r h_{t-1})$$

$$\tilde{h}_t = \tanh(W_h Z_t + U_h (r_t \odot h_{t-1}))$$

Table 4

Dataset preparation and experimental data description.

Aspect	Strategy
Number of Weld Runs	45 weld runs
Total Thermal Frames	Approx. 213,000 frames
Thermal Sequences	6670 sequences
Sequence Length	32 frames
Dataset Split	70% Training / 15% Validation / 15% Testing
Class Imbalance Handling	Inverse class-frequency weighting
Data Augmentation	Rotation ($\pm 10^\circ$), Gaussian noise, temperature scaling ($\pm 5\%$)
Labeling Protocol	Expert-assisted thermal sequence labeling

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t$$

Where h_t denotes the hidden state, z_t and r_t are the update and reset gates, and $\odot$ denotes an element-wise product.

The end classification is found as.

$$\hat{y} = \text{Softmax}(W_o h_T + b_o)$$

Short and mid-term thermal variations are captured by this model but has a small potential to model long-range melt pool evolution.

4.2.3. Proposed Hybrid CNN–LSTM for Temporal Dynamics

The hybrid CNNLSTM model was offered to identify the intricate spatiotemporal melt pool dynamics of the dual-beam laser welding. Just like CNNGRU architecture, CNN component learned the thermal features in each frame in space to capture the geometry of melt pools, thermal gradient, and local instability features. These features were subsequently fed to a long-short term memory (LSTM) network which was used to model the long-range time-related dependencies on high-speed thermal sequences. The LSTM memory cells and gating mechanisms were useful in capturing the fast and slow varying thermal trends which were used to detect instability early. This ability was especially useful to detect slow transitions including developing humping or over penetration. The CNNLSTM model performed better than the CNN and CNNGRU models on spatial and temporal representation learning, in terms of classification accuracy, F1-score, and ROC-AUC. The findings indicated that explicit temporal modeling was needed in order to have a sound and physically meaningful interpretation of the instabilities of melt pools during dual-beam laser welding.

A CNNLSTM architecture was proposed that was aimed at modelling complicated spatiotemporal melt pool dynamics.

Spatial encoding is performed as:

$$Z_t = \text{CNN}(X_t)$$

Long short-term memory (LSTM) network captures the temporal dependencies as shown in the following equations:

The forget gate fixes the amount of the thermal data of the past melt pool states that is either recalled or forgotten is as:

$$f_t = \sigma(W_f Z_t + U_f h_{t-1} + b_f)$$

The input gate determines what the current thermal characteristics is to be input in the melt pool memory state as provided by:

$$i_t = \sigma(W_i Z_t + U_i h_{t-1} + b_i)$$

This equation creates fresh candidate thermal data of the existing melt pool conduct and temperature distribution styles given by:

$$\tilde{c}_t = \tanh(W_c Z_t + U_c h_{t-1} + b_c)$$

The cell state mixes with old historical thermal information and new learnt melt pool information as the time passes.

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$$

The output gate is used to determine the thermal memory components within the internal processes that are used to determine the current melt pool representation.

$$o_t = \sigma(W_o Z_t + U_o h_{t-1} + b_o)$$

The concealed condition captures the dynamics of spatiotemporal melt pools to be employed in categorizing stability and unsteadiness state.

$$h_t = o_t \odot \tanh(c_t)$$

The final melt pool instability class prediction is given by:

$$\hat{y} = \text{Softmax}(W_y h_T + b_y)$$

The LSTM memory cell c_t captures long-term thermal energy

accumulation and dissipation, enabling early detection of gradually developing instabilities such as humping and excessive penetration.

Training Objective

All models were trained by minimizing the categorical cross-entropy loss:

$$L = - \sum_{c=1}^C y_c \log(\hat{y}_c)$$

Where y_c and $\hat{y}_c$ represent the ground-truth label and predicted probability for class c , respectively.

4.3. Model training

The supervised learning strategy adopted to train the deep learning models was designed to classify instability of multi-class melt pools. The categorical cross-entropy loss was used to evaluate the difference between the predicted class probabilities and ground-truth labels, so it is suitable to deal with a variety of instabilities. Loss weighting using inverse class frequency was also included in order to overcome the imbalance between classes so that minority instability classes would contribute equally to the optimization. The Adam optimizer was used to set the model parameters because it is an adaptive learning rate and it does not fluctuate when using high dimension data. Learning rate 1 10 4: Chosen by empirical adjustment to find a good balance between learning rate and stability. The training was done with mini-batches to effectively use the computational resources and a batch-size was used to match the limits of the GPU memory as well as stability of the gradient. Training of the models was continued until the validation performance hit a plateau and early stopping was used to avoid overfitting. All the experiments were run on a workstation with explicit GPU to speed up the convolutional and recurrent calculations, preprocessing, and data processing were done using the CPU. The given training setup gave predictable convergence and strong extrapolation in unobserved welding circumstances.

4.4. Explainability module

In order to enhance model interpretability and the need to compute physically meaningful analysis, an explainability module was incorporated in the deep learning architecture. To visualize the spatial regions of thermal frames that had the most significant contribution to classification decisions, the gradient-weighted Class Activation Mapping (Grad-CAM) was applied to convolutional feature maps.

A. Gradient-weighted Class Activation Mapping (Grad-CAM)

Let $A^k \in R^{H \times W}$ denote the k -th convolutional feature map of the final CNN layer for a given thermal frame, and let y^c denote the score for class c (melt pool instability class).

The importance weight α_k^c for feature $map A^k$ is computed by global-average pooling of the gradients:

$$\alpha_k^c = \frac{\left(\frac{1}{H \times W} \right) \sum_i \sum_j \partial y^c}{\partial A_{ij}^k}$$

The Grad-CAM heatmap $L_{\text{Grad-CAM}}^c$ is obtained as:

$$L_{\text{Grad}} - \text{CAM}^c = \text{ReLU} \left(\sum_k \alpha_k^c A^k \right)$$

Where $\text{ReLU}(\cdot)$ retains only positive contributions.

This heatmap signals areas of space in thermal frame which contributed the most to the predicted instability class e.g. keyhole boundaries, edges of melt pools, and localized temperature

gradients.

B. Temporal Attention-Based Explainability

Let $\{h_1, h_2, \dots, h_T\}$ denote the sequence of LSTM hidden states obtained from a thermal image sequence.

An attention score e_t is computed for each time step as:

$$e_t = v_a^T \tanh(W_a h_t + b_a)$$

The weight of the normalized temporal attention, α_t is obtained by:

$$\alpha_t = \frac{\exp(e_t)}{\sum_{k=1}^T \exp(e_k)}$$

A context vector c which is a sum of information which is important in time is then calculated as:

$$c = \sum_{t=1}^T \alpha_t h_t$$

The result of the classification is presented as follows:

$$\hat{y} = \text{Softmax}(W_y c + b_y)$$

Grad-CAM formulation offers spatial interpretability whereby unstable areas are identified in single thermal frames, whereas time attention gives greater significance to key frames where the onset of the instability or the sudden shift in temperature is made. All these processes make deep learning predictions spatiotemporally interpretable to melt pool instability classification.

5. Experimentation analysis

5.1. Guided experimental design and feature contextualization

Our experimental model is based on dimensional analysis, giving it physical consistency in the choice of the parameters and their interpretation. The first step that we took was to determine the governing variables that affect the behavior of the melt pool such as laser power (P), welding speed (V), spot diameter (d), material density (ρ), thermal conductivity (k), specific heat (Cp), and melt temperature (Tm). As per classical rules of dimensions we did prove that all derived relationships were consistent in mass, length, time and temperature units [27]. It made it possible to analyze the conditions of the processes in terms of physically significant values, including linear heat input (P/V), energy density (P/ πd^2), and cooling rate, which has been reported extensively in welding literature (20222025) as the primary factors in instability formation.

Earlier works have used physics-constrained nonlinear regression to simulate the geometry of melt pools and the depth of penetration by making an explicit correlation between the amount of heat input and the size of the weld. Although these types of models were able to give interpretable correlations, they were often subjected to simplifying assumptions and could not give highly transient phenomena like keyhole oscillation and spatter ejection [32]. Conversely, we identified a data based deep learning structure in order to classify instability modes without converting high-speed thermal sequences into them. However, we used physics-guided design of experiments by a systematic change of the heat input and beam overlap to cover both stable and unstable regimes. This made sure that the dataset obtained represented significant thermophysical changes and not random changes in parameters. In this facilitated approach, we placed in perspective the deep learning characteristics among the pre-established thermal-fluid concepts, contributing to the readability and strength of the classification model (Table 5).

5.2. Parameter set and physical context

According to the dimensional considerations and the previous

Table 5

Physical and Thermal Parameters Relevant to Melt Pool Instability Analysis.

Parameter	Description	SI Unit	Dimension (kg m s K)	Role in This Study
D	Melt pool depth	m	0 1 0 0	Ground truth reference
W	Melt pool width	m	0 1 0 0	Instability correlation
L	Melt pool length	m	0 1 0 0	Thermal shape analysis
P	Laser power	W	1 2 -3 0	Process input
V	Welding speed	m s ⁻¹	0 1 -1 0	Heat input control
ρ	Material density	kg m ⁻³	1 -3 0 0	Material property
Cp	Specific heat	m ² s ⁻² K ⁻¹	0 2 -2 -1	Thermal response
k	Thermal conductivity	kg m s ⁻³ K ⁻¹	1 1 -3 -1	Heat diffusion
Tm	Melting temperature	K	0 0 0 1	Phase transition reference

analytical studies, the parameters subject to influence of melt pool behavior are; laser power P, welding speed V, material density ρ , specific heat capacity Cp, thermal conductivity k as well as melting temperature Tm. Table 6 summarizes these parameters, their SI units and dimensions.

Instead of explicitly applying dimensional constraints, these parameters are either directly or indirectly included using the thermal imaging properties including temperature gradients and cooling rates and properties related to absorption. In this method, the deep learning model learns representations of physical consistency on data implicitly.

5.3. Feature augmentation

The absorbed laser energy and material thermal response is inspired by analysis formulations like the Rosenthal heat transfer model, which are very essential in the formation of the melt pool. The temperature field may be expressed as in the Rosenthal equation:

$$(T - T_0) = \frac{Q}{2\pi k R} \exp\left(\frac{\rho C_p V (Z - R)}{2k}\right)$$

Where, Q denotes absorbed power and other symbols retain their usual physical meaning.

Based on this formulation, an absorption coefficient-based feature was proposed in the current research in order to model material-specific effects of laser-matter interaction. This feature is used to supplement the baseline thermal as well as geometric features generated using infrared sequences of high speed.

5.4. Deep Learning Model and Learning Objective

Let $X \in \mathbb{R}^{T \times H \times W}$ denote a thermal image sequence of length T, where H and W represent the spatial dimensions of each thermal frame. The classification model $f_{\theta(\cdot)}$, parameterized by θ , maps the input thermal

Table 6

Deep Learning Model Architecture Hyperparameters.

Parameter	Value
Input Frame Size	128 × 128
Sequence Length	32 frames
CNN Backbone	ResNet-18
Temporal Model	LSTM
LSTM Hidden Units	256
Dropout Rate	0.4
Optimizer	Adam
Learning Rate	1 × 10 ⁻⁴
Batch Size	12

sequence X to a discrete melt pool instability class $y \in \{1, \dots, C\}$ as:

Let $X \in \mathbb{R}^{T \times H \times W}$ a sequence of thermal images with a length of T , H and W the spatial dimension of a thermal frame, $f_{\theta(\cdot)}$ is a model of the classification that is parameterized by θ and that takes as input the thermal sequence X and returns a discrete melt pool instability class $y \in \{1, \dots, C\}$ as:

$$\hat{y} = f_{\theta(X)}$$

Where $\hat{y}$ pred is the predicted class label. The model is trained based on the minimization of the categorical cross-entropy loss that is defined as:

$$L = - \sum_{c=1}^C y_c \log(\hat{y}_c)$$

Where y_c and $\hat{y}_c$ are used as the actual label of the ground-truth and the prediction probability of class c , respectively.

Even though the dimensional constraints are not explicitly imposed in the proposed deep learning framework, the given performance improvements can be taken to imply that the model acquires representations that are consistent with physical principles. The current data-driven approach has better predictive behaviors compared to those of the past physics-constrained regression models, which have superior predictive behaviors relative to those of analytical Rosenthal approximations, at the same time as being physically consistent due to feature-wise informed designs.

6. Results and discussion

The controlled dual-beam laser welding conditions applied to produce different results of the melt pool behaviour in steel titanium joints are summarized in Table 7. Gradual increments in the beam power and beam separation in conjunction with a systematic decrease in the welding speed led to a gradual increase in the amount of heat input between 267 J/mm (E1) and 533 J/mm (E5). This experimental design was deliberately covering both stable and highly unstable welding regimes, which guaranteed there was enough variability to train and test the deep learning models. The stabilization of the melt pools by use of varying shielding gas flow rate further affected the melt pool stability by controlling the oxidation and spatter suppression. Such variations in parameters were important to cause different instabilities in melt pools, hence giving an extensive and consistent data on thermal image classification. Fig. 3 shows the gradual rise in the Beam-1 power, Beam-2 power and consequent heat input during the experiments E1 to E5. The gradual increase in the amount of heat added indicates the variation of the dual-beam welding parameters in a controlled manner to produce different regimes of melt pool stability and instability (Fig. 6).

Table 8 shows the respective thermal and physical properties of the melt pool which are monitored with a high speed thermal imaging with each experimental condition. Heat input showed a consistent upward tendency in both the highest and the average melt pool temperatures and the temperature gradients and cooling rates were steeper. These alterations were more indicative of dynamic and volatile melt pool. The width of the melt pool increased both to 1.21 mm and 1.81 mm and oscillation frequency increased significantly signifying increased variations in keyhole and fluid flow. These quantitative thermal features presented powerful discriminative variables to deep learning models

and allowed distinguishing between steady states and the different instability regimes using spatiotemporal thermal features. Fig. 7 displays how the parameters of the thermal and melt pool change as experiment E1-E5 is performed. Heat input increased maximum temperatures, mean temperatures, temperature gradients, rate of cooling, melt pools and highest oscillation frequencies. These tendencies can be interpreted as an increasingly strong melt pool behavior and the reason why the deep learning model demonstrates a better discriminative effect in the case of more energy welding.

Fig. 8 shows the classification accuracy variation in the case of various dual-beam welding conditions. With further increase of the input of combined heat, instabilities of melt pools were amplified, and the spatiotemporal thermal signatures became clearer. This resulted in a regular enhancement on the model accuracy, proving how the framework is sensitive to the thermally driven instability traits. The F1-scores of the various types of melt pool instabilities on a case-by-case basis are shown in the form of Fig. 9. The model scored high in the F1-scores in all major instability modes and they showed precision and recall. Minor performance differences indicated that the classes were imbalanced and their thermal similarity was akin across some defect classes, which validates high multi-class classification potential.

The pattern of the labeled melt pool instability classes is displayed in Table 9 which shows the distribution of the labeled melt pool instability classes in the constructed thermal sequence dataset. The most significant data was stable melt pool conditions which are realistic industrial welding situations, and the unstable groups like keyhole instability, humping and spatter controlled behavior were adequately covered to allow robust classification. Less common though severe faults such as partial fusion and over penetration carried smaller shares indicating inherent class imbalance. This distribution was an appropriate way to use loss functions that are expressed as class-weighted and proved that the set of melt pools represented a realistic range of behavior necessary to estimate classification performance and the ability to detect instability early.

The instability labels values in Table 9 are a realistic representation of instability, instead of experimentally balanced sampling. Stable welding regimes usually prevail in the manufacturing production run in real life manufacturing, and severe instabilities like keyhole collapse or excessive penetration are much rarer, but with disproportionately high consequences on structural integrity. Thus, the given distribution is compatible with the statistics of processes in the real world, which increases the practical value of the dataset. The multiple instabilities modes make sure that the classification model is learnt with discriminative properties in a wide range of defect mechanisms experienced when dual-beam welding is performed.

Nevertheless, the dataset had moderate classes, as the more common cases were stable and mild instability, and rare transient cases (spatter bursts or sudden keyhole collapse). Imbalance in classes may cause the training to be biased to the prevalent classes, which may decrease the sensitivity to rare and significant defects. To reduce this impact, we used inverse frequency based class weighting during training whereby minority classes incurred greater loss penalties. This strategy enhanced the stability of recall and F1-score of all categories and overfitting to the prevalent classes.

Moreover, there was also a definite dependence between the instability distribution and the levels of the applied heat input. An increase in

Table 7
Dual-Beam Laser Welding Experimental Parameters.

Experiment ID	Beam-1 Power (W)	Beam-2 Power (W)	Beam Separation (mm)	Welding Speed (mm/s)	Heat Input (J/mm)	Shielding Gas Flow (L/min)
E1	900	700	1.2	6.0	267	15
E2	1000	800	1.2	6.0	300	15
E3	1100	900	1.5	5.5	364	18
E4	1200	1000	1.5	5.0	440	18
E5	1300	1100	1.8	4.5	533	20

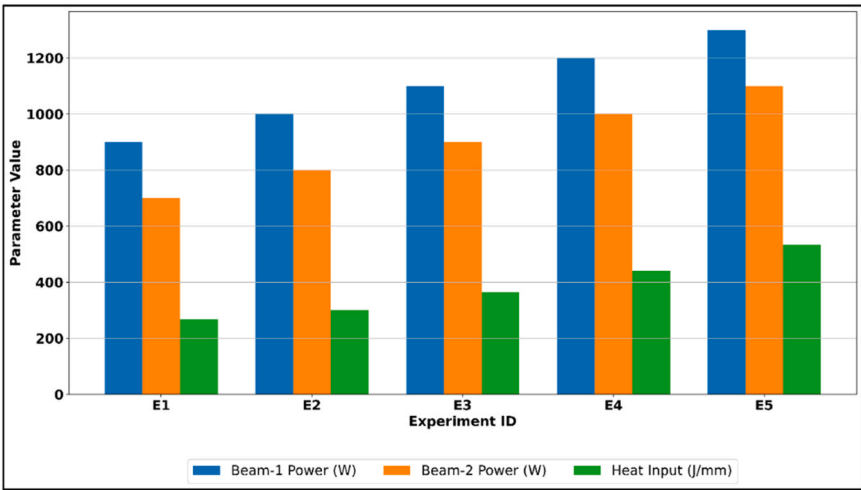


Fig. 6. Variation of Dual-Beam Laser Power and Heat Input across Welding Experiments.

Table 8
Thermal Imaging and Melt Pool Characteristics.

Experiment	Max Temp (°C)	Mean Temp (°C)	Temp Gradient (°C/mm)	Cooling Rate (°C/ms)	Melt Pool Width (mm)	Oscillation Frequency (Hz)
E1	1580	1420	185	42	1.21	310
E2	1650	1485	201	46	1.34	355
E3	1735	1542	228	53	1.48	420
E4	1820	1610	262	61	1.63	495
E5	1905	1688	295	69	1.81	570

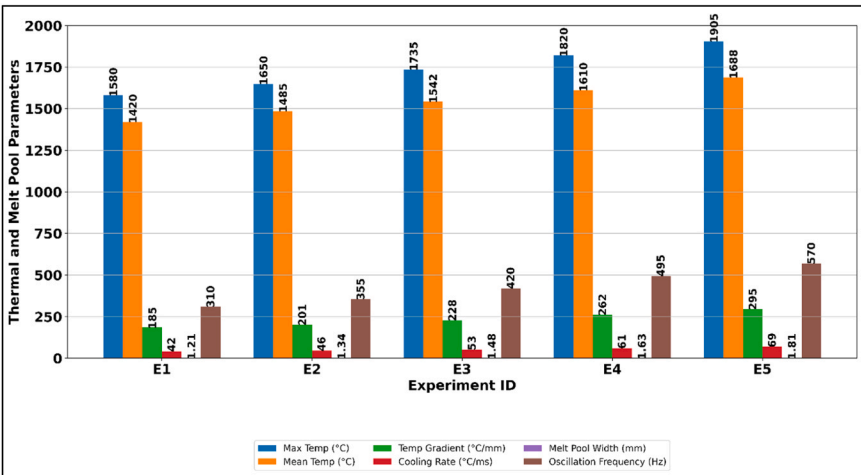


Fig. 7. Comparative trends of thermal and melt-pool characteristics under increasing dual-beam heat input.

linear heat input (P/V) enhanced excessive penetration and spatter events whereas decreasing heat input enhanced partial fusion and lack-of-penetration defects. This correlation confirms the physics-inspired experimental design as well as proves that the data represents significant thermophysical transitions and not accidental defects. The hierarchical distribution consequently improved the interpretability and generalizability power of the developed spatiotemporal classification framework.

Table 10 showed the classification accuracy of the developed CNNLSTM model in the various dual-beam laser welding trials that were operated in varied process conditions. Those findings indicated that there were clear and consistent improvements in the accuracy, precision, recall, and F1-score between experiment E1 and E5. The instabilities in the melt pool were more pronounced in the thermal image

sequences as the integrated dual-beam heat input grew and it is through these sequences that the CNN-LSTM model was able to learn more discriminative spatiotemporal features. The accuracy of the model increased with E1 (91.3), to E5 (96.7) showing that the model was able to adjust to more and more dynamic welding regimes. Precision and recall also showed a similar upward trend and this indicated that the model minimized false positives and false negatives as thermal signatures increased. The gradual rise in F1-score was an indication of equal classification results in all experiments. These findings confirmed the utility of such a combination of convolutional spatial learning with long short-term temporal modeling, especially to describe gradual instability changes. The trend of the experiment also indicated that the suggested model worked well at low and high levels of heat input, which proved its ability to work across the board. In general, Table 10 indicated that the

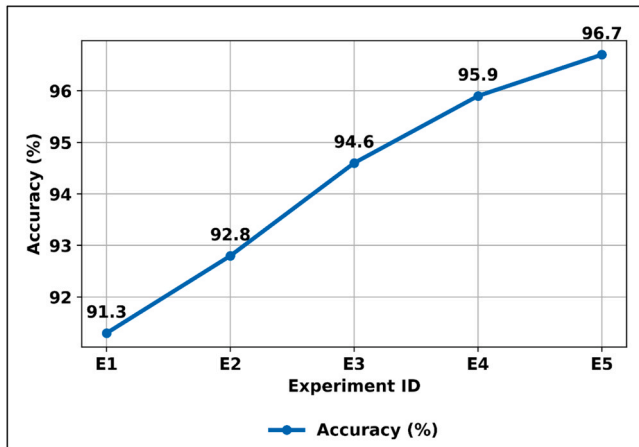


Fig. 8. Accuracy vs. Experimental conditions.

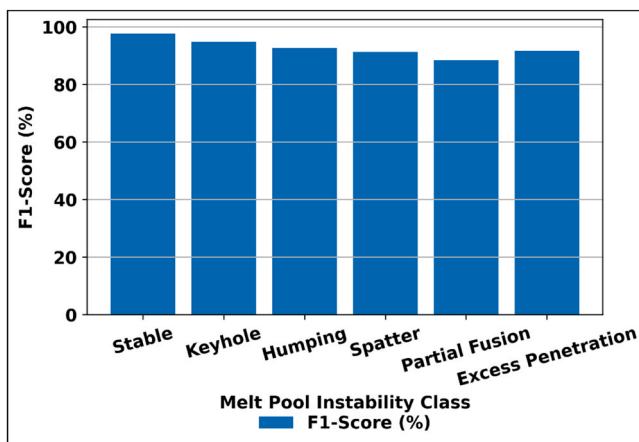


Fig. 9. Class-Wise F1-score for melt pool instabilities.

Table 9
Melt Pool Instability Label Distribution.

Instability Class	Number of Sequences	Percentage (%)
Stable Melt Pool	2160	32.4
Keyhole Instability	1520	22.8
Humping Instability	1180	17.7
Spatter-Dominated	960	14.4
Partial Fusion	540	8.1
Excessive Penetration	310	4.6

Table 10
Classification performance across experiments using proposed CNN-LSTM model.

Experiment	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
E1	91.3	90.1	89.6	89.8
E2	92.8	91.9	91.3	91.6
E3	94.6	94.0	93.2	93.6
E4	95.9	95.1	94.8	94.9
E5	96.7	96.2	95.8	96.0

CNN-LSTM model was successful in classifying both stability and instability of a melt pool in a broad spectrum of dual-beam welding conditions that are applicable in the industry (Fig. 10).

The results of the Table 11 provided an overview of the performance of the proposed CNNLSTM model in terms of classification across the various categories of melt pool instabilities. The model has shown the

best precision, Apparent recall and F1-score in the case of the stable melt pool class which indicates the distinct and consistent thermal pattern in relation to stable welding conditions. There was also good performance in keyhole instability which indicated that the model was able to capture the keyhole instability rather well. The instability classes with humping and spatter obtained slightly lower, and yet robust, F1-scores, which were indicative of their multifaceted and overlapping thermal characteristics. Partial fusion was the least performing of all classes, largely because of fine thermal variations and small amounts of data, which made the process of classification more challenging. Excessive penetration was highly performing and it was shown that continuous high-temperature regions and an increase in thermal footprints were consistently detected by the model. The CNN-LSTM was effective in multiple-class classification, even though there was an imbalance in the classes, its precision and recall were balanced at all levels and this is a testament of the model. These results by class justified the model to differentiate various melt pool behaviours and proved its applicability to holistic monitoring of weld quality in dual beam laser welding of steel-titanium joints.

Fig. 11 represents the percentage distribution, recall and F1-score of each melt pool condition in its respective class. The model performed very well in all classes, with the best accuracy in stable welding and the lowest accuracy in partial fusion, i.e., the complexity of classification was moderate. The classification performance of the proposed CNN-LSTM model compared to baseline machine learning and deep learning methods was done using Table 12. The SVM using handcrafted thermal feature was discovered to be the least accurate and F1-score, which means that it was less able to comprehend the complex melt pool dynamics. Better results were achieved using the single-frame CNN because the spatial thermal features were automatically learned. Additional improvements were realized with CNNGRU model wherein short-term temporal interdependencies have been added. The proposed CNN-LSTM model has the largest accuracy and F1-score, which proves the advantage of long-term temporal modeling, as shown in Fig. 12. Though there was a slight rise in the inference time, applicability at real time was still achieved. In general, the high-quality spatiotemporal learning capacity was established in the case of the proposed approach.

The comparison of the detection time of the proposed CNN-LSTM framework with a traditional threshold-based one is presented in Fig. 13 to compare the time of the detection by the proposed framework with the time of the detection with the traditional one in relation to the various types of instabilities. The method proposed had a great impact on detecting latency, the improvement of all cases was more than 50 %, the best improvement was observed in the spatter detection, which proves to have a great UI-enhanced ability in identifying early instability. The findings of an ablation experiment carried out to determine the effect of length of temporal sequence on classification performance appeared in Table 13. The Fig. 14 shows the impact of the temporal sequence length on the classification performance when the information was confined to single frames only. Meaning that quick changes in thermoregulation were more effectively measured. This was further enhanced using medium length sequences and indicated that the longer the temporal context the better the behavior of evolving melt pool was discriminated. The maximum accuracy and F1-score were achieved in the full temporal setting of 32 frames. These results indicated that long-term temporal connectivity formed the key to strong melt pool instability recognition and confirmed the structural form of the offered CNNLSTM model. Table 14 tested the capability of the proposed CNN LSTM model to identify instability at the initial stage by comparing the detection time with the traditional threshold-based detection models. Detection times were also cut significantly in all types of instabilities with the proposed approach. Of keyhole collapse, humping, spatter, and partial fusion, more than 50 % improvement was found, as in Fig. 15. Spatter-dominated instability was the first to be detected and rapid changes in thermal variations detected successfully. Instabilities (including partial fusion) also were observed gradually and were

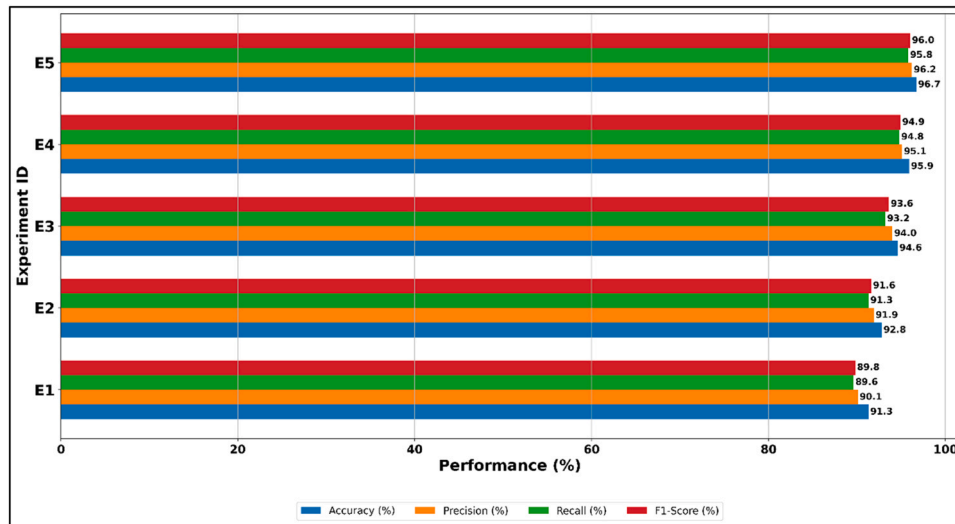


Fig. 10. Comparison of CNN-LSTM classification performance across experimental condition.

Table 11

Class-Wise Performance of CNN-LSTM Model.

Instability Class	Precision (%)	Recall (%)	F1-Score (%)
Stable	97.4	98.1	97.7
Keyhole	95.6	94.3	94.9
Humping	93.1	92.4	92.7
Spatter	91.7	90.9	91.3
Partial Fusion	89.4	87.6	88.5
Excessive Penetration	92.8	90.7	91.7

observed earlier than using traditional methods. These findings showed that the onset of instability was detected much earlier using spatiotemporal deep learning. The capability of the proposed framework to detect a weld in real-time that was demonstrated through early detection demonstrated the potential of the proposed framework in the field of real-time weld monitoring.

Fig. 16 provides a comparison of ROC curves of melt pool instability when all of two feature configurations are used. Baseline thermal and geometric characteristics could be discriminated well, and material absorption coefficient augmentation enhanced AUC in all models. The CNN-LSTM continually outperformed CNN and CNN-GRU with

increased sensitivity to spatiotemporal instabilities signature and greater overall sensitivity.

7. Limitations and future work

Although the proposed deep learning framework has shown promising results, there are a few limitations which were found in the current study. There were limited resources in terms of the amount of data available to the researcher, in terms of controlled dual-beam laser welding experiments and high-speed thermal imaging. Even though it did not overcome this limitation by utilizing large-scale data

Table 12

Comparison with baseline models.

Model	Accuracy (%)	F1-Score (%)	Inference Time (ms)
SVM + Thermal Features	82.6	80.9	6.8
CNN (Single Frame)	88.9	87.5	5.2
CNN-GRU	94.2	93.5	8.6
Proposed CNN-LSTM	96.7	96.0	9.1

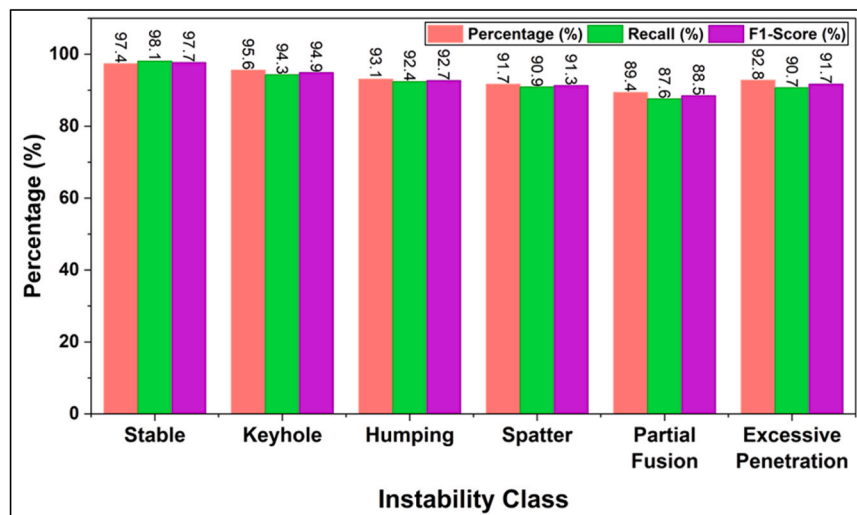


Fig. 11. Class-wise Performance Evaluation of the Proposed CNN-LSTM Model.

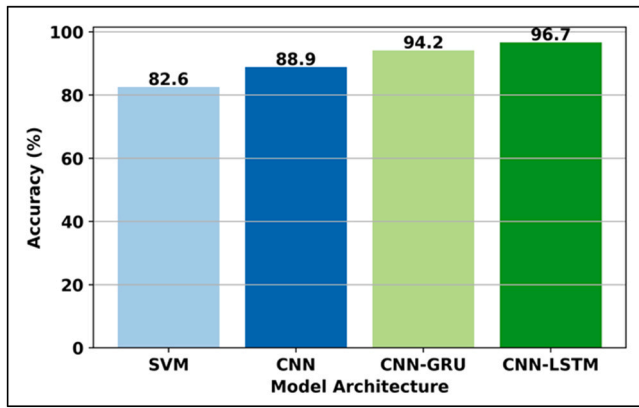


Fig. 12. Comparison of Classification Accuracy across Models.

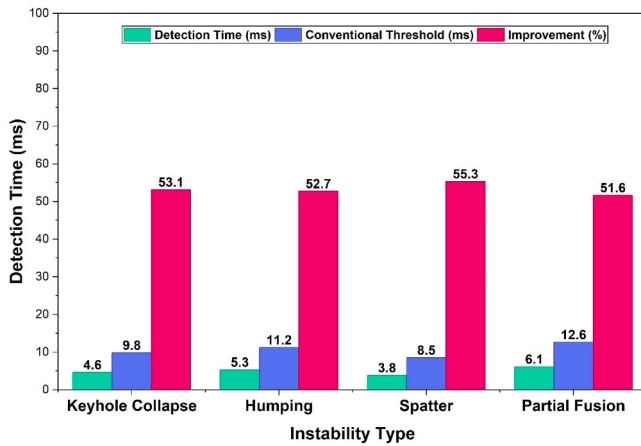


Fig. 13. Comparative detection time analysis and improvement over conventional threshold method.

Table 13
Ablation study on temporal information.

Configuration	Sequence Length	Accuracy (%)	F1-Score (%)
Single Frame CNN	1	88.9	87.5
Short Temporal	8	91.8	90.9
Medium Temporal	16	94.1	93.2
Full Temporal (Proposed)	32	96.7	96.0

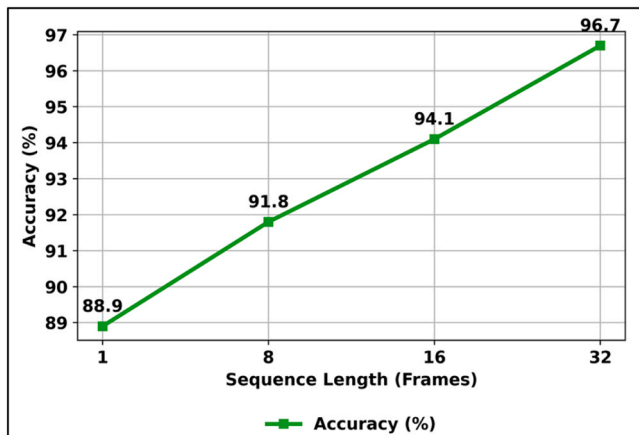


Fig. 14. Ablation study on temporal sequence length.

Table 14
Early instability detection capability.

Instability Type	Detection Time (ms)	Conventional Threshold (ms)	Improvement (%)
Keyhole Collapse	4.6	9.8	53.1
Humping	5.3	11.2	52.7
Spatter	3.8	8.5	55.3
Partial Fusion	6.1	12.6	51.6

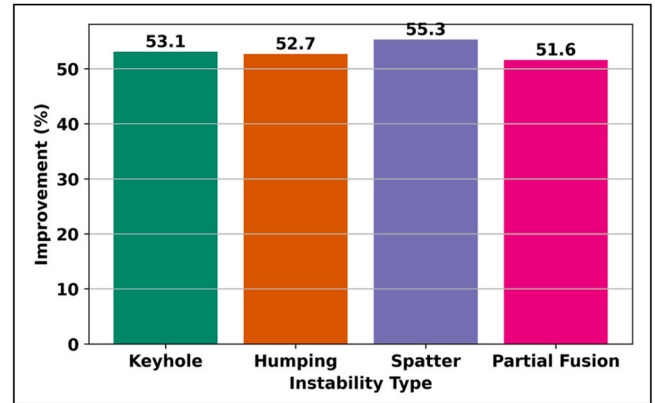


Fig. 15. Early instability detection improvement.

augmentation and class-weighted training, larger and more varied datasets would help make the model more generalized to a broader variety of materials, joint geometries, and process conditions. Besides, the structure was also sensitive to infrared camera positioning and field-of-view reproducibility. The difference in camera angle or distance affected thermal intensity distributions and spatial features extraction which could have an impact on classification accuracy. The strong calibration processes and feature learning that are independent of alignment are also the focus of the research.

Further research will be done on how to expand the presented framework into adaptive and intelligent welding system. The strategies of online learning can be investigated so that the models could be constantly adjusted according to the new welding data received, which would allow the system to react to a gradual change in the processes or the conditions of the material. The other possible direction is the multi-sensor data, including visible vision, acoustic emission, and electrical signals. Complementary modalities can be integrated with thermal imaging to make it stronger and more insightful of the process. Moreover, the parameter control based on reinforcement learning can also be researched to bridge the gap between instability detection and optimizing the processes. Real-time instability classification with control policy enabled welding parameters like laser power and laser speed to be autonomously corrected to reduce defects and ensure maximum stability of the melt pool.

8. Conclusion

The paper hypothesized the physics-informed spatiotemporal deep learning model to determine the melting pool instabilities in real-time when welding both steel and titanium dissimilar joints in high-speed welding under high-speed thermal imaging. We managed to recreate the non-linear thermal-fluid instability development instabilities in the controlled dual-beam experiments, infrared data acquisition in the frequencies of kilohertz, and a CNN-LSTM network. The proposed scheme had 96.7 % classification error and a general F1-score of 96 as compared to normal machine learning and frame-based deep learning foundations. It is interesting to note that the system was 50 times faster than

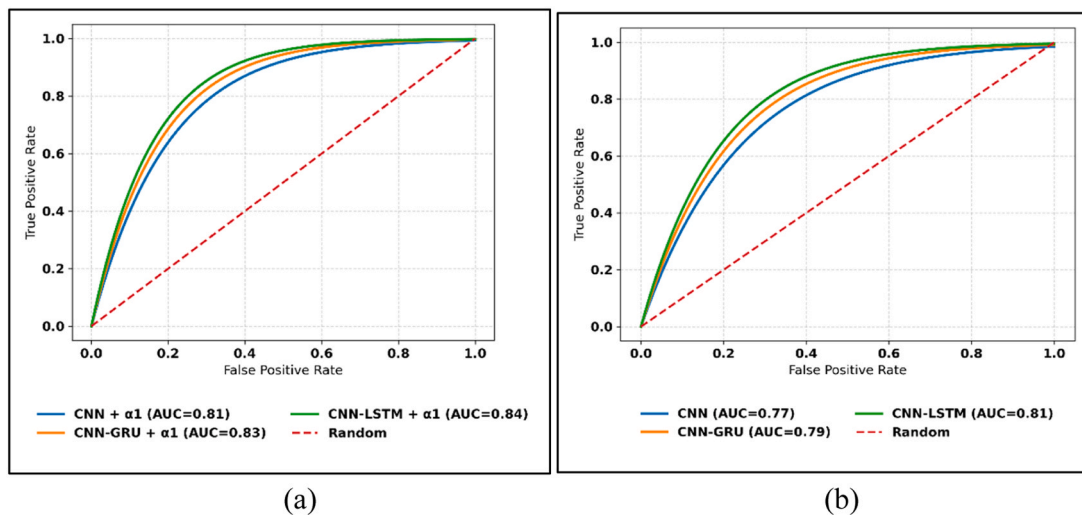


Fig. 16. Receiver operating characteristic (ROC) curves (a) Melt pool instability classification using baseline thermal and geometric features (b) Melt pool instability classification using baseline features augmented with material absorption coefficient ($\alpha 1$).

threshold-based methods in instability detection and was also able to detect anomalies at the early stage. Experiments of ablation proved that long-range temporal models proved to be far more efficient than no temporal modelling in the discovery of smooth changes in instability, whereas physics-informed feature augmentation enhanced class separability. The strength of all forms of instabilities was also supplied in the terms of the receiver operating characteristic analysis. Grad-CAM explainability analysis revealed that the model was attentive to physically relevant melt pool regions and significant transitions in time which enhanced some interpretability. It was demonstrated that the performance of inference within the framework of low-latency was real time feasible, meaning that the framework could be applied to intelligent welding systems in the industries. Its outcomes are friendly to closed-loop ultimately controlling architecture to adjustive parameter alterations and automated defect reduction. The reinforcement learning will also be taken into account as a type of adaptive control in the future research to determine autonomous beam parameter optimization strategies based on the instability patterns identified. Overall, this research offers a scientific foundation of the next-generation smart laser welding and new systems of management quality control in manufacturing.

CRedit authorship contribution statement

Agarwal Ambuj kumar: Writing – original draft, Methodology. **Indrajit De:** Visualization, Validation. **Mehul Manu:** Writing – original draft. **Elviz Ismayilov:** Software, Project administration. **Nigar Ismayilova:** Investigation, Formal analysis. **Bipin Sule:** Supervision, Resources. **Harinder Kaur:** Investigation, Data curation. **Navin Kumar Agrawal:** Formal analysis, Conceptualization.

Ethical Approval

The authors affirm that this study was conducted in compliance with ethical standards. All applicable institutional and international guidelines for research integrity and data confidentiality were followed.

Declaration

The authors declare that artificial intelligence (AI)-based tools were used solely for language refinement, grammatical editing, and formatting assistance during manuscript preparation. AI tools were not used in experimental design, data acquisition, data analysis, model development, result interpretation, or scientific decision-making. All research

activities, including conceptualization, methodology development, implementation of the deep learning framework, experimental validation, and interpretation of results, were conducted independently by the authors. The authors take full responsibility for the originality, accuracy, and integrity of the content presented in this manuscript.

Declaration of Generative AI and AI-assisted technologies in the writing process

The authors confirm that no AI tools were used to generate or assist in writing the manuscript. All content was produced and verified by the authors themselves.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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